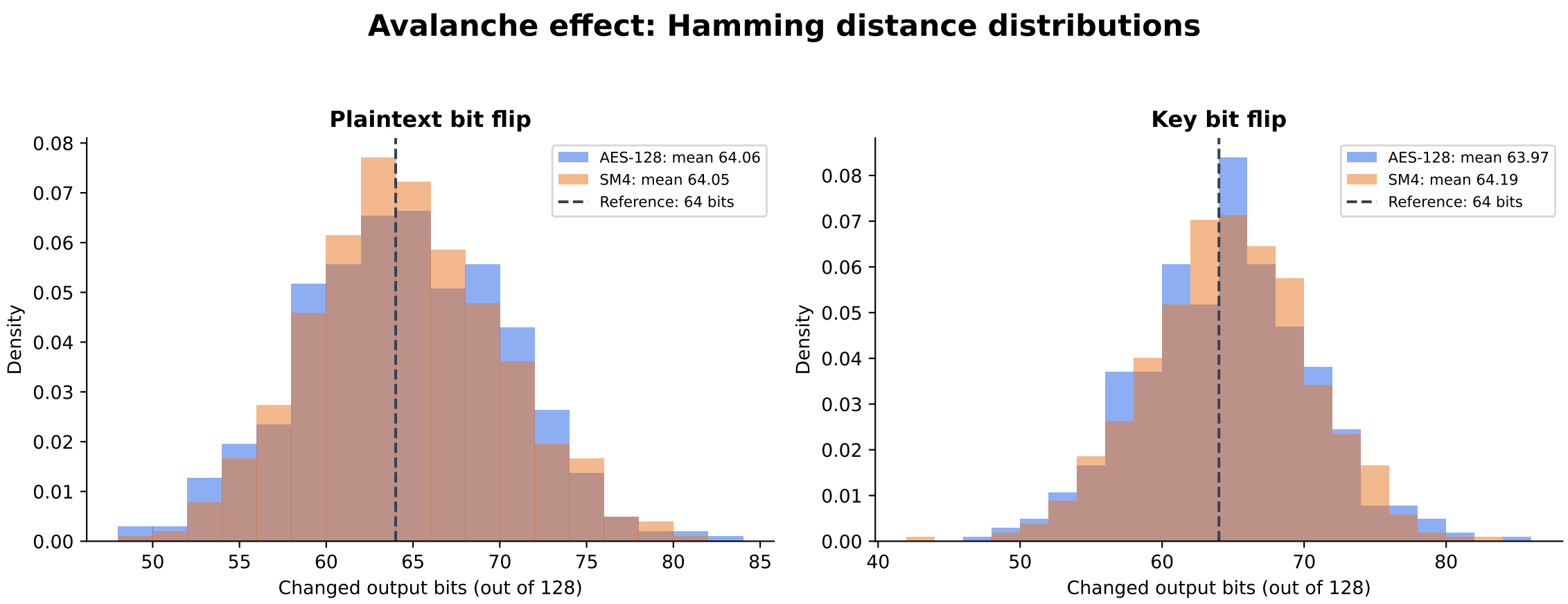
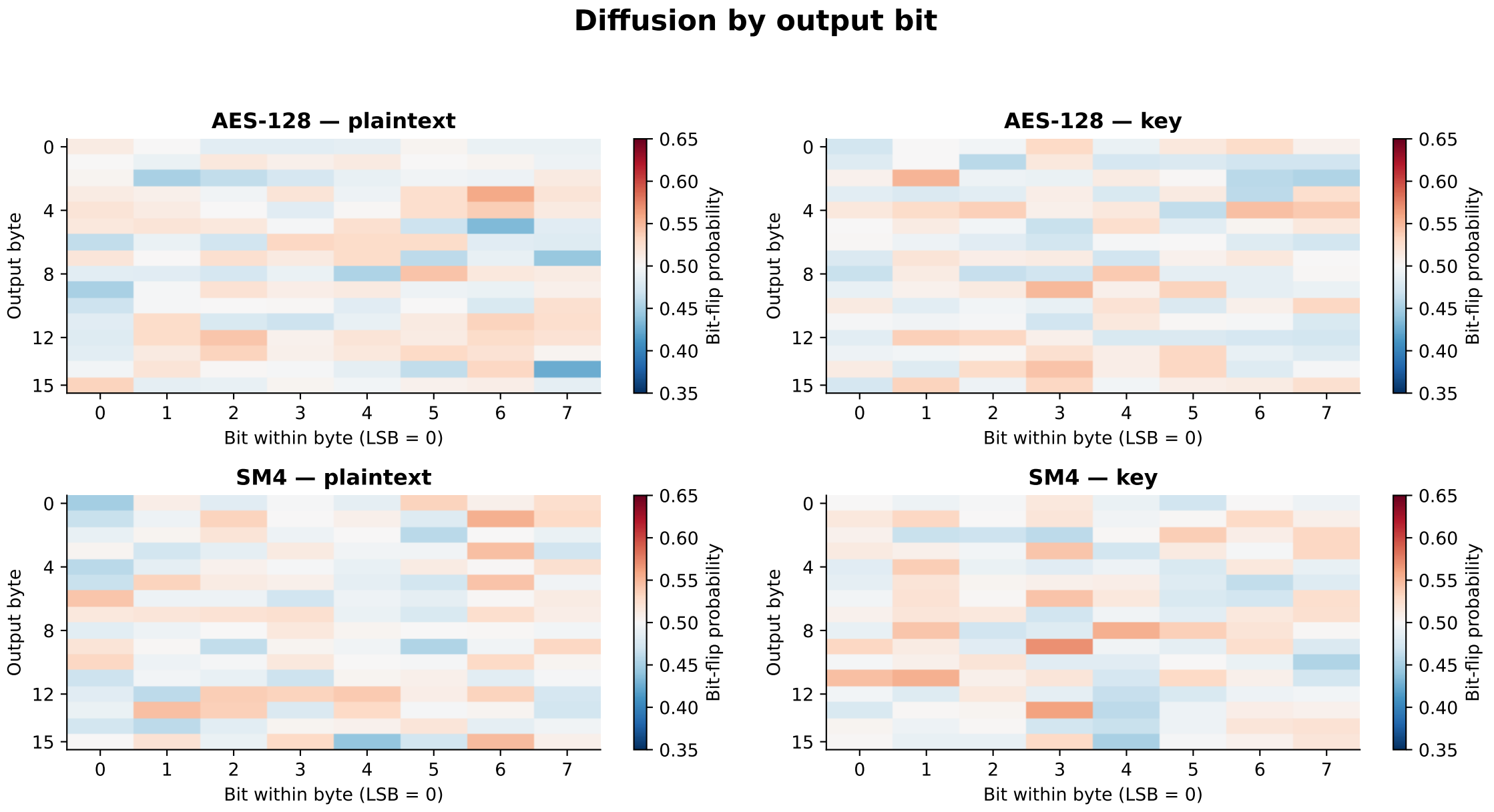
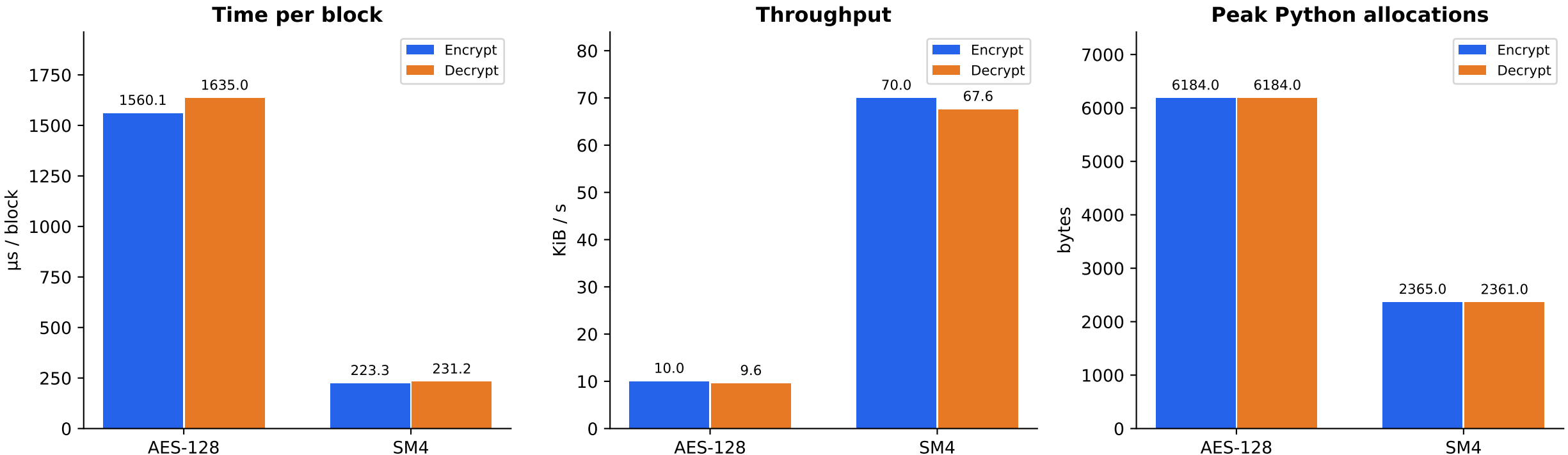
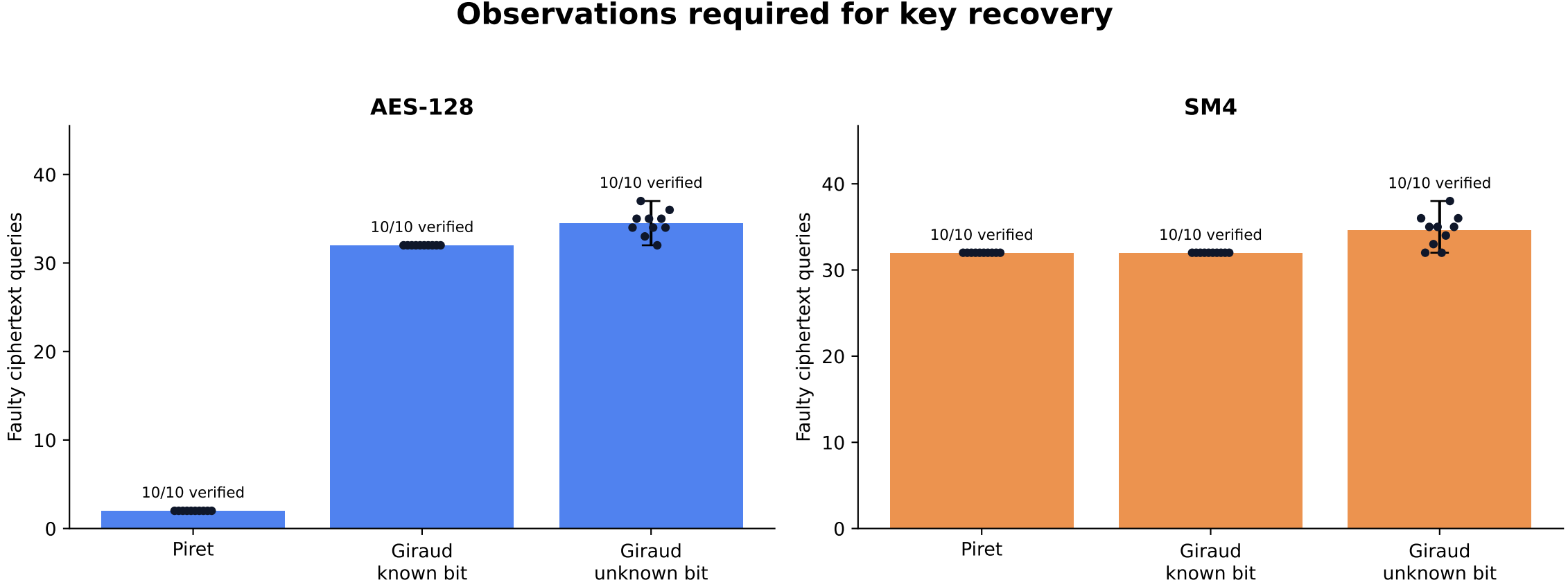

Avalanche effect: Hamming distance distributions
Plaintext bit flip
Key bit flip
AES-128: mean 64.06
SM4: mean 64.05
Reference: 64 bits
AES-128: mean 63.97
SM4: mean 64.19
Reference: 64 bits
Density
Changed output bits (out of 128)
One bit flipped at a time; 50% reference. Finite sample, not a security certification.


Diffusion by output bit
AES-128 — plaintext
AES-128 — key
SM4 — plaintext
SM4 — key
Output byte
Bit within byte (LSB = 0)
Bit-flip probability
0
4
8
12
15
0
1
2
3
4
5
6
7
0.65
0.60
0.55
0.50
0.45
0.40
0.35
Probabilities pooled across input bits and trials; not a full SAC matrix.

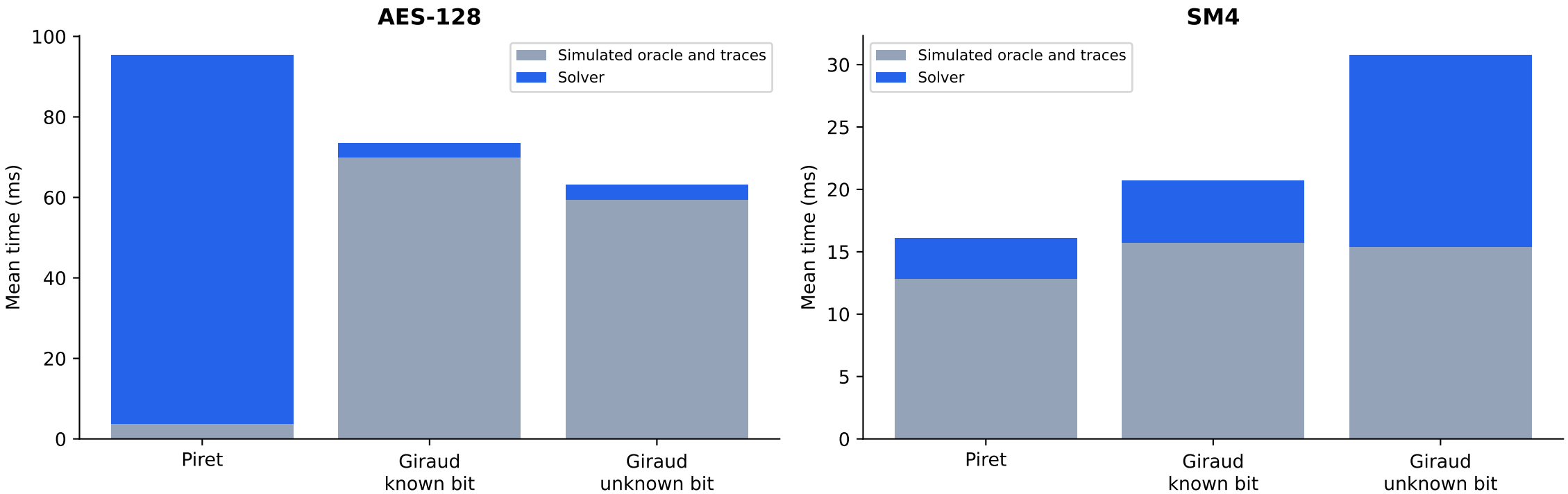
Python implementation performance



Key expansion included. Time: median of 5 batches. Memory measured separately; not process RSS.

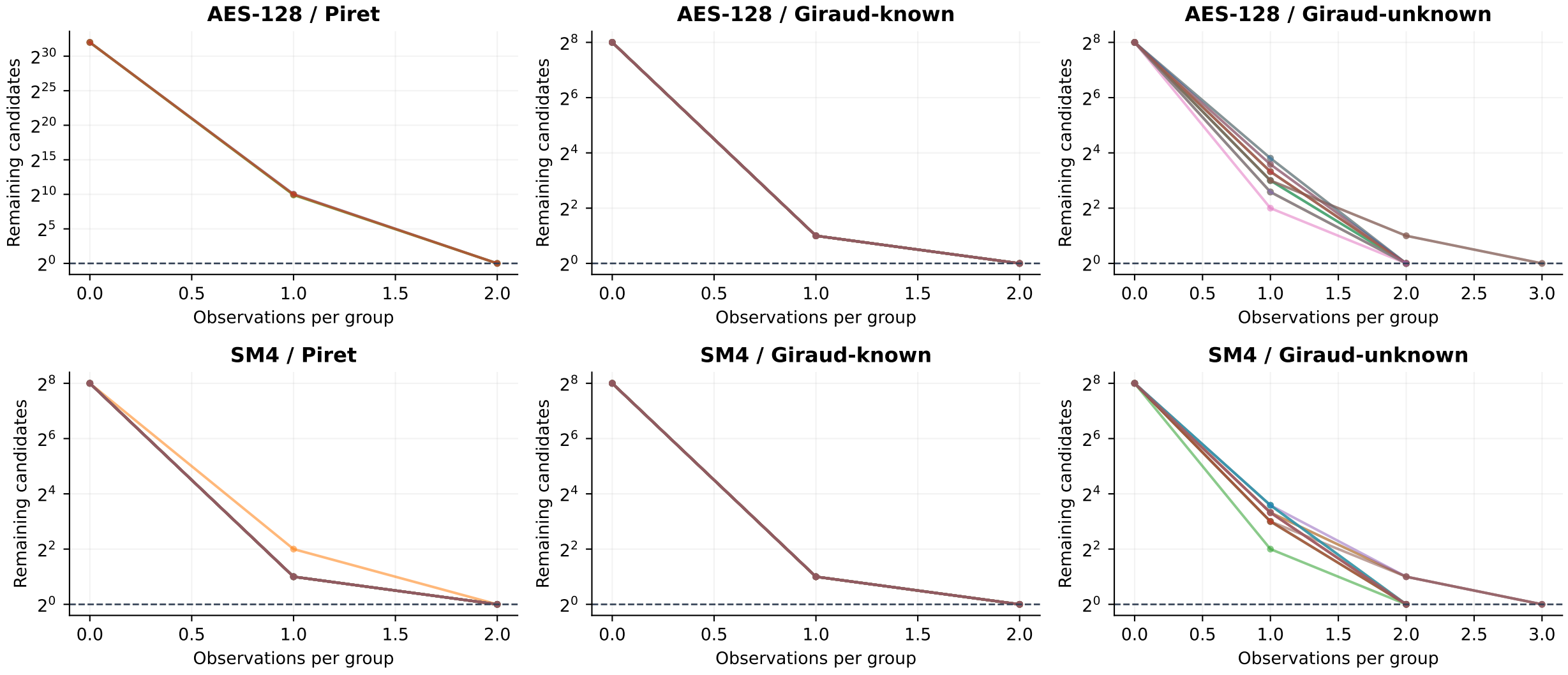

Observations required for key recovery
AES-128
SM4
Faulty ciphertext queries
0
10
20
30
40
10/10 verified
10/10 verified
10/10 verified
10/10 verified
10/10 verified
10/10 verified
Piret
Giraud
known bit
Giraud
unknown bit
Piret
Giraud
known bit
Giraud
unknown bit
Bars: mean; whiskers: min–max; dots: trials. One additional correct ciphertext per trial.

Attack time: simulation and solving

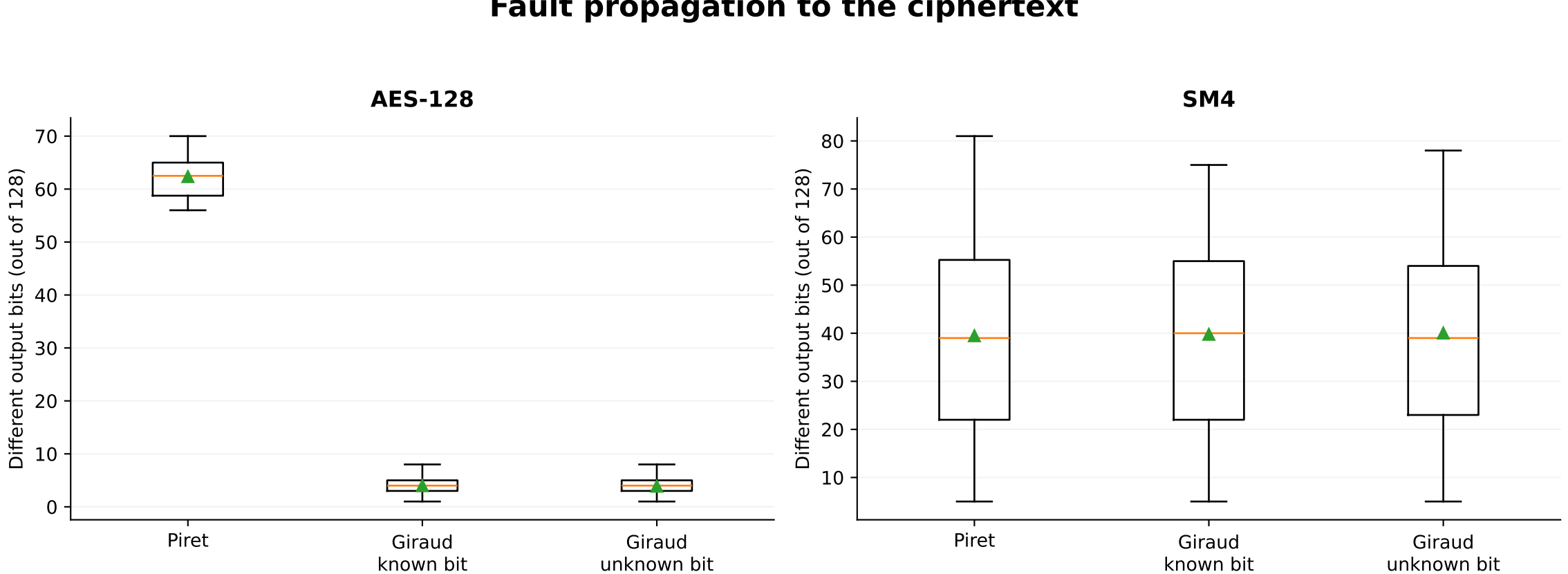


Software timings do not estimate the duration of a physical fault-injection campaign.

Candidate reduction during the attack



First trial of each model. Each curve tracks one key byte; AES Piret tracks a four-byte group.


Fault propagation to the ciphertext
AES-128
SM4
Different output bits (out of 128)
Different output bits (out of 128)
0
10
20
30
40
50
60
70
10
20
30
40
50
60
70
80
Piret
Giraud
known bit
Giraud
unknown bit
Piret
Giraud
known bit
Giraud
unknown bit
Distribution across observations; propagation depends on the round and injection point of each model.